Swayam Bansal

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Work Experience

Community Dreams Foundation — Remote

AI Engineer — *May 2026 – Present*

- Built and deployed an AI-powered HR email automation system handling 50+ emails/day, separating high-frequency FAQ requests into deterministic template responses and routing context-specific questions through retrieval to improve accuracy while reducing unnecessary LLM usage.
- Added production guardrails for sensitive-content detection, retrieval failure, and hallucination control, automatically escalating harassment-related or unsupported requests to HR; introduced an LLM-as-judge evaluation layer to improve response-quality scoring and support safer automated replies.
- Built a real-time AI voice assistant for routine HR calls, integrating telephony, speech recognition, LLM reasoning, and text-to-speech into an end-to-end conversational system that resolved 60%+ of routine calls before human escalation.
- Redesigned a production RAG assistant into a multi-agent architecture, separating project-specific, training, and general-support workflows; introduced semantic retrieval and role-aware SQL authorization checks so project agents could retrieve only information accessible to each user, preventing cross-project data exposure.
- Routed long-form training conversations to a smaller, lower-cost model to reduce inference cost, while unresolved support requests automatically generated HR tickets for human follow-up instead of returning unsupported answers.

De-Bias — Remote

Data Analyst Intern — *Feb 2025 – June 2025*

- Unified multi-year Pennsylvania housing-assistance survey data into a consistent analytical dataset, reconciling changing survey questions, categorical fields, and messy free-text responses to make previously fragmented data usable for modeling.
- Built and evaluated predictive models to identify the strongest indicators associated with housing-funding outcomes, using feature selection and imbalance-aware modeling to support more data-driven prioritization of limited government assistance and surface potential allocation inefficiencies.

Penn State University — University Park, PA

Math Tutor — *Aug 2023 - Dec 2024*

- Led 50+ tutoring sessions in calculus, linear algebra, and other advanced mathematics, translating complex quantitative concepts into structured problem-solving strategies.

Pennsylvania State University — Harrisburg, PA

Research Assistant — *May 2023 - Aug 2023*

- Redesigned a legacy particle-collision system for CFD simulations, evaluating collision approximations that traded physical realism for computational efficiency and enabling near-real-time simulation of large particle sets.

Project Experience

AgentPay SDK — Payment Infrastructure for AI Agents | https://github.com/Swayam-Bansal/AgentPay-SDK

- Designed a rail-agnostic payment SDK for AI agents, supporting agent wallets, policy-controlled transfers, escrow, and idempotent payment intents so autonomous agents can transact under configurable safety constraints rather than unrestricted execution.
- Built an immutable double-entry ledger and policy layer with allowlists, transaction limits, and pause controls, exposing the system through both local and HTTP interfaces and validating payment, escrow, and end-to-end flows with automated tests.

Keyzz — Computer Vision Air Piano | https://github.com/Swayam-Bansal/Keyzz

- Built a real-time vision system that turns a printed paper keyboard into an interactive piano, using webcam-based hand tracking and key detection to trigger audio without specialized hardware.
- Designed automatic calibration and stabilization to handle camera positioning and noisy finger tracking, then added synchronized audio playback and a rhythm-game mode to turn the vision pipeline into a usable interactive product.

Education

Pennsylvania State University — University Park, PA

Bachelor of Science in Computer Science with Minor in Applied Mathematics — *Dec, 2025 | GPA: 3.50*

Skills

Languages: Python, SQL, Java, C/C++
AI/ML: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, Sentence-BERT, OpenCV, MediaPipe
Tools/Platforms: ChromaDB, Supabase, PostgreSQL, Git, GitHub Actions, Docker, AWS/GCP, FastAPI